

11.17 GAUSS'S LAW

According to this law, the total electric flux through a closed surface is equal to the charge (in Coulomb) enclosed by that surface. If $\vec{D}$ represents the electric flux density through an elementary surface area $\vec{ds}$ drawn about any point, then the flux through this elementary area is

$$d\phi = \vec{D} \cdot \vec{ds}$$

Thus, the total flux through a closed surface is

$$\phi = \int d\phi = \oint_S \vec{D} \cdot \vec{ds}$$

$\therefore$ According to the Gauss's law

$$\phi = \oint \vec{D} \cdot \vec{ds} = q \text{ in SI system, where } q \text{ is charge}$$

and for free space, the above equation can be written as

$$\phi = \oint \epsilon_0 \vec{E} \cdot \vec{ds} = q$$

11.17.1 Gaussian Surface

An imaginary closed surface of any shape drawn in an electric field for the purpose of solving problems concerning electric flux is called *Gaussian surface*. The shape of the Gaussian surface is chosen on the basis of symmetry of the problem, so that the expression for Gauss's law,

$$\phi = \oint \vec{D} \cdot \vec{ds}, \text{ can be evaluated conveniently.}$$

11.17.2 Proof of Gauss's Law

Suppose a charge q , assumed to be situated at the origin of the co-ordinate axes as shown in Fig. 11.8. Let S be a Gaussian surface around it. Consider an elementary area $\vec{ds}$ at $\vec{r}$. The electric flux through $\vec{ds}$ is given by

$$\phi = \vec{D} \cdot \vec{ds}$$

where $\vec{D}$ is electric flux density at $\vec{r}$.

and

$$\phi = \int d\phi = \oiint \vec{D} \cdot \vec{ds}$$

But the electric displacement vector for free space is

$$\vec{D} = \epsilon_0 \vec{E} \quad (\text{in SI})$$

$\therefore$

$$\vec{D} = \epsilon_0 \frac{q}{4\pi\epsilon_0 r^2} \hat{r} = \frac{q}{4\pi r^2} \hat{r}$$

Hence, we have

$$\phi = \oiint \frac{q}{4\pi r^2} \hat{r} \cdot \vec{ds} = \frac{q}{4\pi} \oiint \frac{\hat{r} \cdot \vec{ds}}{r^2} \quad \dots(12)$$

But

$$\oiint \frac{\hat{r} \cdot \vec{ds}}{r^2} \int d\Omega = 4\pi$$

i.e., the solid angle subtended by a closed surface on a point inside it is 4π

$$\therefore \phi = \frac{q}{4\pi} 4\pi = q$$

Which is Gauss's Law, i.e., the total electric flux in SI system through a closed surface is equal to the charge (in Coulomb) enclosed by the surface.

From eqn. (12), it is clear that the electric flux ϕ over a surface is equal to

$$\phi = \frac{q}{4\pi} \Omega$$

With $\Omega = \oiint \frac{\hat{r} \cdot \vec{ds}}{r^2}$ as the solid angle

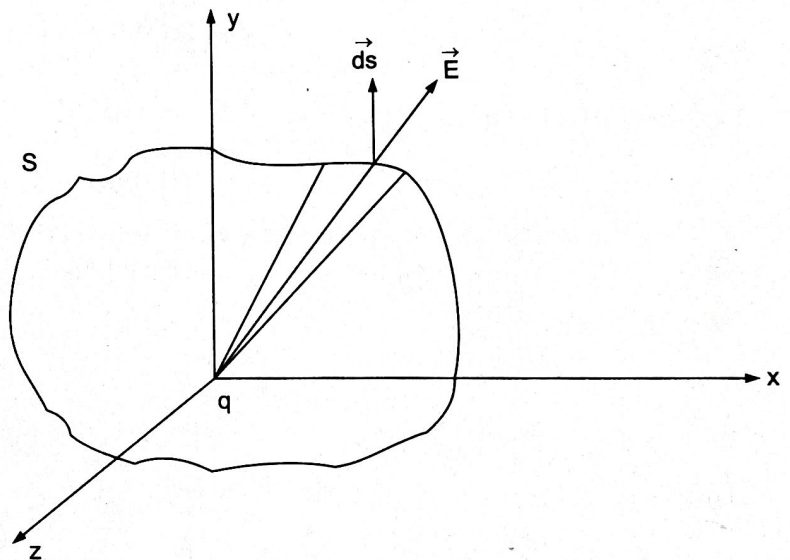


Fig. 11.8 Depiction of Gaussian surface

subtended by the surface at the location of point charge. If the charge enclosed is zero, the total flux through the surface is zero. It implies that either no lines of electric force are cutting the surface or the number of lines of force entering the surface is equal to the number of lines of force coming out of it.

If the Gaussian surface encloses more than one charge then the Gauss's law can be written as

$$\phi = \oiint \vec{D} \cdot \vec{ds} = \sum_{i=1}^n q_i \quad \dots(13)$$

If there is continuous distribution of charge, then we have

$$\phi = \oiint \vec{D} \cdot \vec{ds} = \int dq = \iiint \rho dv \quad \dots(14)$$

where ρ is the volume density of charge.

If the Gaussian surface can be divided into a large number of surfaces $S_1, S_2, S_3, \dots$, etc., then the expression for Gauss's Law can be written as

$$\phi = \oiint \vec{D} \cdot \vec{ds} = \iint_{S_1} \vec{D}_1 \cdot \vec{ds}_1 + \iint_{S_2} \vec{D}_1 \cdot \vec{ds}_2 + \iint_{S_3} \vec{D}_1 \cdot \vec{ds}_3 + \dots$$

11.17.3 Differential Form of Gauss's Law

Gauss's law can also be written as

$$\oiint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} \quad (\because \vec{D} = \epsilon_0 \vec{E}) \quad \dots(15)$$

From Gauss's divergence theorem, we have

$$\oiint \vec{E} \cdot d\vec{s} = \iiint_v \text{div} \cdot \vec{E} dv \quad \dots(16)$$

Combining eqns. (15) and (16), we have

$$\iiint_v \text{div} \cdot \vec{E} dv = \frac{q}{\epsilon_0} \quad \dots(17)$$

If q is the total charge contained in the volume distribution having volume density of charge ρ , then we have

$$q = \iiint_v \rho dv \quad \dots(18)$$

From eqns. (17) and (18)

$$\iiint_v \text{div} \cdot \vec{E} dv = \frac{\iiint_v \rho dv}{\epsilon_0} \quad \dots(19)$$

Which gives

$$\text{div} \vec{E} = \frac{q}{\epsilon_0}$$

But

$$\text{div} \vec{E} = \nabla \cdot$$

$\therefore$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \dots(20)$$

or

$$\vec{\nabla} \cdot \vec{D} = \rho \quad \dots(21)$$

the above equations have been derived from Gauss's law. $\vec{\nabla} \cdot \vec{E}$ expresses the emergence of electric flux from a point where volume density of charge is ρ . These equations express the Gauss's theorem in differential form, because $\vec{\nabla}$ is differential operator.

11.19 APPLICATION OF GAUSS'S LAW

Gauss's law can be used to calculate the electric field strength if the symmetry of the charge distribution is high. Here we will select the Gaussian surfaces which are having symmetries about the charge distribution. The method has been illustrated for the following problems.

11.19.1 Line of Charge

Figure 11.9 shows the section of an infinite rod of charge, the linear charge density λ (charge per unit length) being constant for all points on the line. Let the line charge is lying parallel to x -axis. Consider a point P at a distance r from the linear charge at which the electric field strength has to be calculated. We choose a circular cylinder of radius r and length l as a Gaussian surface, closed at each end normal to the axis. The Gaussian surface can be divided into three sections S_1 , where P is lying and S_2 and S_3 , the surfaces of the ends of Gaussian cylinder. The total charge enclosed by the Gaussian surface is $q = \lambda l$.

Now applying Gauss's law, we get

$$\oiint \epsilon_0 \vec{E} \cdot \vec{ds} = q$$

or
$$\epsilon_0 \left(\iint_{S_1} \vec{E}_1 \cdot \vec{ds}_1 + \iint_{S_2} \vec{E}_2 \cdot \vec{ds}_2 + \iint_{S_3} \vec{E}_3 \cdot \vec{ds}_3 \right) = q$$

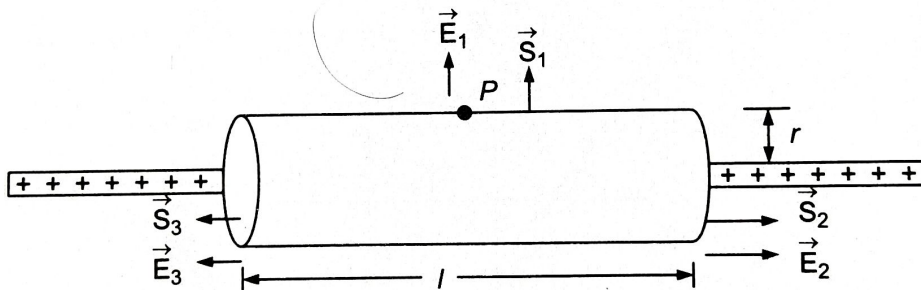


Fig. 11.9 An infinite rod of charge, showing a cylindrical Gaussian surface

Because of symmetry $\vec{E}_2 = \vec{E}_3 = 0$

and let $\vec{E}_1 = \vec{E}$, then

$$\epsilon_0 \iint_{S_1} \vec{E} \cdot \vec{ds} = q$$

or
$$\epsilon_0 \iint_{S_1} \vec{E} \cdot \vec{ds} = \lambda l$$

Since E is constant at the surface and $\vec{E}$ and $\vec{S}_1$ are parallel everywhere, we have

$$\epsilon_0 \iint_{S_1} \vec{E} \cdot d\vec{s}_1 = \epsilon_0 E \iint_{S_1} ds_1$$

or

$$\epsilon_0 E \iint_{S_1} ds_1 = \lambda l$$

But

$$\iint_{S_1} ds_1 = 2\pi r l,$$

$\therefore$

$$\epsilon_0 E \cdot 2\pi r l = \lambda l$$

or

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

which is radially directed along the cylinder.

11.19.2 A Sheet of Charge

Figure 11.10 shows a portion of thin, non-conducting, infinite sheet of charge, the surface charge density σ (charge per unit area) being constant. Consider a point P at a distance r from the surface of the sheet. Now draw a cylindrical surface having its upper end through P and lower end at distance r from the sheet. Let S_1, S_3 be the surface areas of the ends of the cylinder and S_2 the surface area of the cylindrical surface. From symmetry, E points at right angles to the ends of the cylinder and away from the plane. Since E does not pierce the cylindrical surface, there is no contribution to the flux from this source. Thus, according to Gauss's law, we have

$$\oiint \epsilon_0 \vec{E} \cdot d\vec{s} = S$$

$$\text{or } \iint_{S_1} \vec{E}_1 \cdot d\vec{s}_1 + \iint_{S_2} \vec{E}_2 \cdot d\vec{s}_2 + \iint_{S_3} \vec{E}_3 \cdot d\vec{s}_3 = \frac{\sigma S}{\epsilon_0}$$

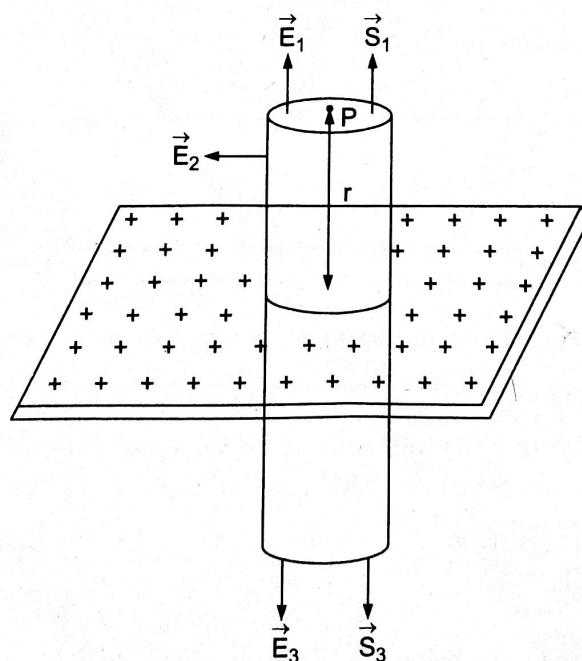


Fig. 11.10 An infinite sheet of charge pierced by a cylindrical Gaussian surface. The cross-section of the cylinder need not be circular

Since

$$\vec{E}_2 = 0, \vec{E}_1 = -\vec{E}_3 = \vec{E} \text{ (say)}$$

and

$$\vec{ds}_1 = -\vec{ds}_3 = \vec{ds} \text{ (say)}$$

Also

$$\vec{E}_1 \parallel \vec{ds}_1, \vec{E}_3 \parallel \vec{ds}_3$$

We have

$$\iint_{S_2} \vec{E}_2 \cdot \vec{ds} = 0,$$

$$\iint_{S_1} \vec{E}_1 \cdot \vec{ds}_1 = \iint_S \vec{E} \cdot \vec{ds} = E \iint_S ds = ES$$

and

$$\iint_{S_3} \vec{E}_3 \cdot \vec{ds}_3 = \iint_S (-\vec{E}) \cdot (\vec{ds}) = E \iint_S ds = ES$$

$\therefore$

$$2ES = \frac{\sigma S}{2\epsilon_0}$$

or

$$E = \frac{\sigma}{2\epsilon_0}$$

Thus, E is same for all points on each side of the plane.

11.19.3 Electric Field Due to Infinite Parallel Sheets of Charge

Let P_1 and P_2 be two infinite plane parallel sheets carrying opposite uniform charge densities (see Fig. 11.11). Let a point P be lying in the region between two charged plates. Now in order to calculate the electric field at point P , draw a normal cylinder of cross-sectional area S through point P having one of its ends in the plate. Using Gauss's Law, we have

$$\oint_S \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$$

Let σ be the surface charge density, then

$$q = \sigma S$$

$\therefore$ Above equation becomes

$$ES = \frac{\sigma S}{\epsilon_0} \quad (\because \vec{E} \text{ and } \vec{ds} \text{ are parallel})$$

or

$$E = \frac{\sigma}{\epsilon_0}$$

which implies that the field between two plates is uniform and independent of the separation between them.

Again consider a point Q lying extreme right of the two plates. Now draw a cylinder of cross sectional area S through this point Q with one of its face lying on plate P_1 . Therefore, from Gauss's Law, we have

$$\oint_S \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$$

Since the charge inside the Gaussian surface is zero i.e., $q = \sigma S - \sigma S$, therefore we have

$$\oint_S \vec{E} \cdot \vec{ds} = 0$$

$$\begin{aligned} \text{or} \quad & ES = 0 \\ \Rightarrow & E = 0 \quad (\because \vec{E} \parallel d\vec{s}) \end{aligned}$$

which clearly indicates that the electric field intensity due to two oppositely charged infinite plane sheets at a point outside the plates is zero.

We can also verify these results pictorially as follows: We know that the electric field intensity at any point at a distance r due to an infinite plane sheet of charge having charge density σ is $\frac{\sigma}{2\epsilon_0}$. Therefore, the left plate (in Fig. 11.11) produces a field $\frac{\sigma}{2\epsilon_0}$ which points away from it as shown in Fig. 11.12) to the left in region I and to the right in region II and III. The right plate produces a field $\frac{\sigma}{2\epsilon_0}$ which points towards it to the right in regions I and II and to the left in region III. The two fields cancel in regions I and II but they conspire in region III. Thus, between the plates the electric field intensity is $\frac{\sigma}{\epsilon_0}$ and outside is zero.

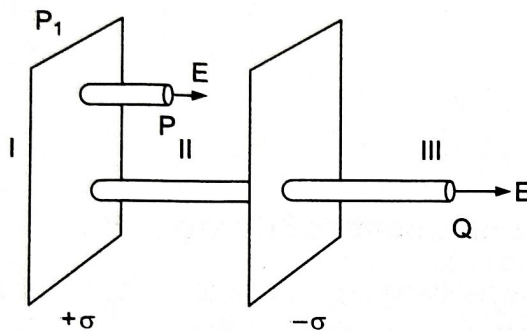


Fig. 11.11

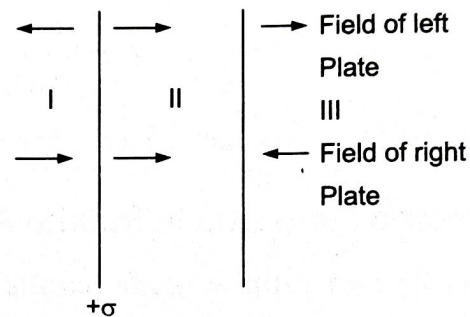


Fig. 11.12

11.19.4 Spherical Charge Distribution

Consider a sphere of charge Q having volume density ρ . Let r be the radius of the sphere. Let P and P' be two points outside and inside the sphere respectively as shown in Fig. 11.13. To calculate the electric field strength at P and P' , we consider Gaussian surfaces S_1 and S_2 respectively through P and P' with O as their centre. Let us consider $OP = r_0$ and $OP' = r_i$. Now apply Gauss's law to the outer Gaussian surface through P , we have

$$\oint \epsilon_0 \vec{E}_0 \cdot d\vec{s}_1 = Q,$$

where $\vec{E}_0$ is the electric field at point P , and $Q = \frac{4}{3}\pi r^3 \rho$

$$\text{or} \quad \epsilon_0 \oint \vec{E}_0 \cdot d\vec{s}_1 = Q$$

$$\text{or} \quad \epsilon_0 \vec{E}_0 \oint d\vec{s}_1 = Q$$

$$\text{or} \quad \epsilon_0 \vec{E}_0 4\pi r_0^2 = Q$$

$$\text{or} \quad E_0 = \frac{Q}{4\pi\epsilon_0 r_0^2}$$

Thus, it is clear that the sphere of charge behaves as if the whole charge is concentrated at the center.

Again apply Gauss's law to the inner Gaussian surface through P' , we have

$$\oiint \epsilon_0 \vec{E}_i \cdot \vec{ds}_2 = Q'$$

where $\vec{E}_i$ is the electric field at point P' and Q' is the charge enclosed by the Gaussian surface S_2 . Therefore,

$$Q' = \frac{4}{3} \pi r_i^3 \rho$$

or

$$Q' = \frac{4}{3} \pi r_i^3 \frac{Q}{\frac{4}{3} \pi r^3} = Q \frac{r_i^3}{r^3}$$

$\therefore$

$$\oiint \epsilon_0 \vec{E}_i \cdot \vec{ds}_2 = Q' = Q \frac{r_i^3}{r^3}$$

or

$$\epsilon_0 E_i \oiint ds_2 = Q \frac{r_i^3}{r^3}$$

or

$$\epsilon_0 E_i 4\pi r_i^2 = Q \frac{r_i^3}{r^3}$$

or

$$E_i = \frac{Q}{4\pi\epsilon_0} \frac{r_i}{r^3}$$

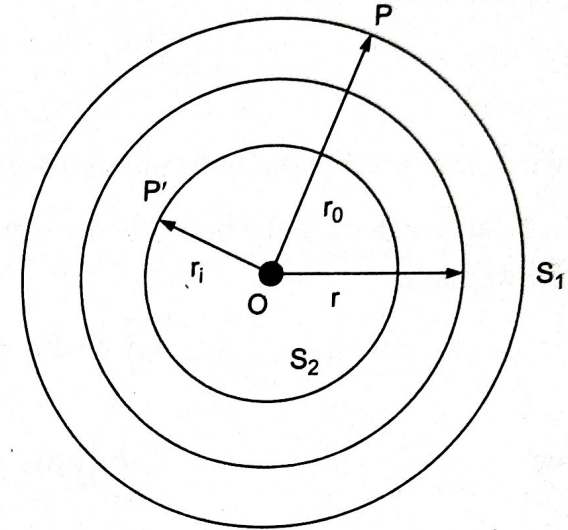


Fig. 11.13 A Spherically symmetric charge distribution, showing two Gaussian surfaces

The above relation shown that the intensity of the electric field inside the sphere of charge varies directly as the distance from the centre. Suppose $r_i = r$, then the electric field due to sphere of charge Q will be

$$E = \frac{Q}{4\pi\epsilon_0} \frac{1}{r^2}$$

which represents the electric field due to sphere of radius r just on the surface of the sphere.

11.19.5 Uniformly Charged Spherical Shell

Let us consider three points P_1 , P_2 and P_3 respectively outside, inside and on the surface of the uniformly charged spherical shell (see Fig 11.14). the corresponding Gaussian surface through these points are S_1 , S_2 and S_3 . Let O be the centre of the shell and $OP_1 = r_0$, $OP_2 = r_i$ and $OP_3 = r$. Let $\vec{E}_0$, $\vec{E}_i$ and $\vec{E}$ be the electric field intensities at P_1 , P_2 and P_3 respectively. Suppose the surface density of charge be σ , then the total charge on the shell will be $Q = 4\pi r^2 \sigma$.

Now, apply Gauss's law through Gaussian surface S_1 , we have

$$\oiint \vec{E}_0 \cdot \vec{ds}_1 = \frac{Q}{\epsilon_0}$$

or

$$E_0 \oiint \vec{ds}_1 = \frac{Q}{\epsilon_0}$$

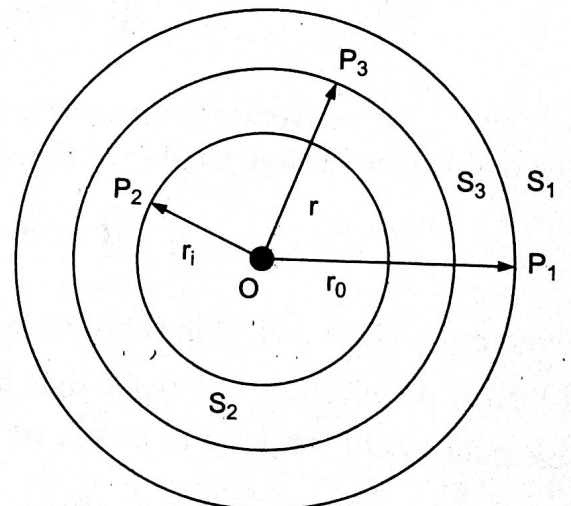


Fig. 11.14 A spherically symmetric charge distribution, showing two Gaussian surfaces

($\because \vec{E}_0$ is having same magnitude everywhere on the surface S_1 and $\vec{E}_0$ is parallel to $\vec{ds}_1$)

$$\text{or} \quad E_0 4\pi\epsilon_0 r_0^2 = \frac{Q}{\epsilon_0}$$

$$\text{or} \quad E_0 = \frac{Q}{4\pi\epsilon_0 r_0^2}$$

Similarly, we can apply Gauss's law through Gaussian surface S_2 , and thus, we have

$$\oiint \vec{E}_i \cdot \vec{ds}_2 = 0, \text{ because charge enclosed is zero.}$$

$$\therefore E_i = 0$$

which implies that the intensity of electric fields inside a shell is everywhere zero.

Now to calculate the intensity of electric field $\vec{E}$ on the surface to the shell, apply Gauss's law through S_3 , we have

$$\oiint \vec{E} \cdot \vec{ds}_3 = \frac{q}{\epsilon_0}$$

$$\text{or} \quad E \oiint ds_3 = \frac{q}{\epsilon_0} \quad (\because \vec{E} \parallel \vec{ds}_3)$$

$$\text{or} \quad E \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$\text{or} \quad E = \frac{q}{4\pi\epsilon_0 r^2}$$

which gives the electric field strength due to uniformly charged shell on the surface of the shell.

11.20 GAUSS'S DIVERGENCE THEOREM

It states that the surface integral of the normal component of electric field vector $\vec{E}$ over the bounding surface S of a volume v is equal to the volume integral of the div $\vec{E}$ over the same volume, i.e.,

$$\oiint \vec{E} \cdot \vec{ds} = \iiint_V \text{div } \vec{E} dv$$

The Gauss's divergence theorem is applicable to all types of vector fields. Let the bounding surface be divided into a large number of elementary surfaces (see Fig. 11.15), say N , then

$$\oiint \vec{E} \cdot \vec{ds} = \sum_{i=1}^{i=N} \iint_{S_i} \vec{E} \cdot \vec{ds}_i$$

where $\vec{ds}_i$ is the area of the i th surface.

Now divide and multiply the right hand side of the above equation by volume v_i , the volume of i th element whose surface area is $\vec{ds}_i$, we have

$$\sum_{i=1}^{i=N} \left(\iint_{S_i} \frac{\vec{E} \cdot \vec{ds}_i}{v_i} \right) v_i$$

Let us consider the limiting case, *i.e.*, as N approaches infinity, v_i also approaches infinity. Therefore, the above equation becomes

$$\sum_{i=1}^{\infty} \text{Lt}_{v_i \rightarrow \infty} \left(\oiint_{S_i} \frac{\vec{E} \cdot \vec{ds}_i}{v_i} \right) v_i$$

But, we know that

$$\text{Lt}_{v_i \rightarrow \infty} \oiint_{S_i} \frac{\vec{E} \cdot \vec{ds}_i}{v_i} = \text{div } \vec{E}$$

Therefore, we have

$$\sum_{i=1}^{\infty} (\text{div } \vec{E}) v_i$$

or

$$\iiint_v \text{div } \vec{E} dv$$

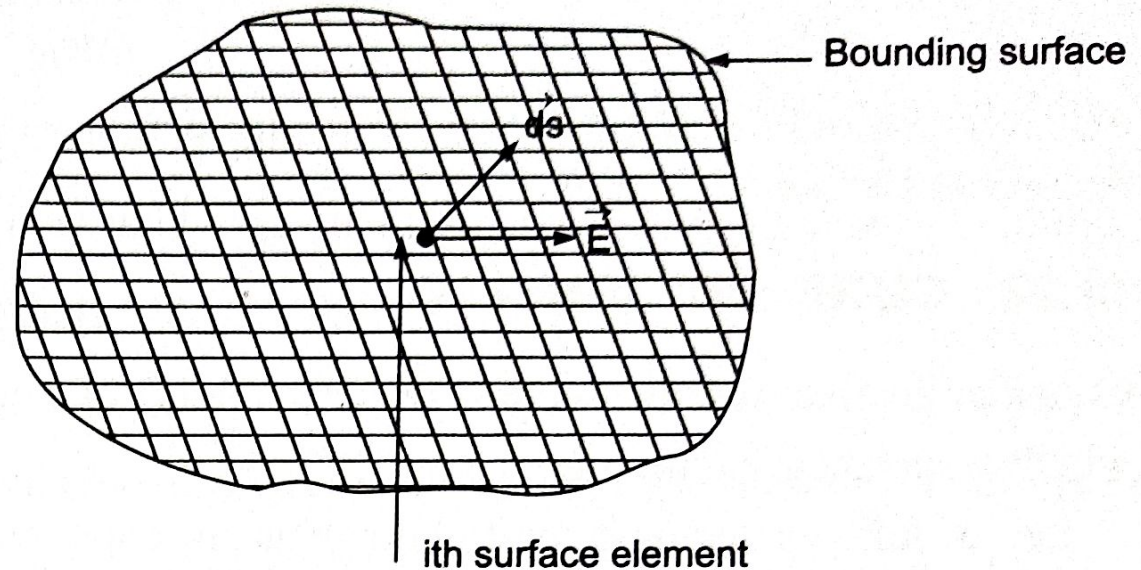


Fig. 11.15 A large surface bounding many elementary surfaces

where we have used the concept that the summation over all the volume elements reduces to volume integral.

$$\therefore \sum_{i=1}^{i=N} \left(\iint_{S_i} \vec{E} \cdot \vec{ds}_i \right) = \iiint_v \text{div } \vec{E} dv$$

or

$$\iint \vec{E} \cdot \vec{ds} = \iiint_v \text{div } \vec{E} dv$$

which is the Gauss's divergence theorem.

11.23 CONTINUITY EQUATION

We know from eqn. (23) that the current density j and current I are related as

$$I = \iint_S \vec{j} \cdot \vec{ds} \quad \dots(27)$$

The magnitude and direction of current are same everywhere and remain unchanged with time for the steady current. But, if the current is not steady, both current and current density vary from point to point as well as from time to time. Thus, we may express current as well as current density as

$$I \equiv I(x, y, z, t) \quad \dots(28)$$

and

$$\vec{j} \equiv \vec{j}(x, y, z, t)$$

Let V be a volume of the current carrying conductor (Fig. 11.19) and ρ be the volume charge density, then total charge in the volume V is given by

$$q = \iiint_V \rho dV \quad \dots(29)$$

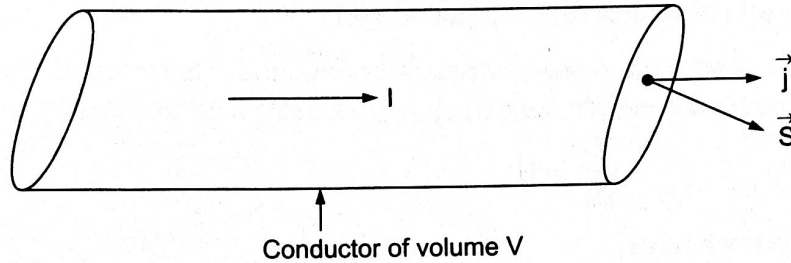


Fig. 11.19 A current carrying Conductor with volume V

Since current is defined as the rate of flow of charge, the current through the conducting cylinder is

$$I = -\frac{dq}{dt} = -\frac{d}{dt} \left[\iiint_V \rho dV \right]$$

where -ve sign signifies the outflow of charge from the cylindrical surface.

Also

$$I = \oiint \vec{j} \cdot \vec{ds}$$

Therefore, from last two equations we have

$$\oiint \vec{j} \cdot \vec{ds} = -\frac{d}{dt} \left[\iiint_V \rho dV \right] \quad \dots(30)$$

Also according to the Gauss's divergence theorem

$$\oiint \vec{j} \cdot \vec{ds} = \iiint_V \text{div} \cdot \vec{j} dV \quad \dots(31)$$

On combining eqns. (30) and (31), we have

$$\iiint_V \text{div} \cdot \vec{j} dV = -\frac{d}{dt} \iiint_V \rho dV = \iiint_V -\frac{d\rho}{dt} dV \quad \dots(32)$$

or

$$\text{Div} \vec{j} = -\frac{d\rho}{dt}$$

or

$$\vec{\nabla} \cdot \vec{j} = -\frac{d\rho}{dt} \quad \dots(33)$$

Equation (33) is known as equation of continuity and represents the physical fact of conservation for charge.

11.23.1 Discussion of Continuity Equation

1. If the current is steady, *i.e.*, its magnitude is not changing with space or times, then the total charge inside the elementary volume remains constant with time and hence

$$\frac{d\rho}{dt} = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{j} = 0$$

Thus for steady currents, $\text{div } \vec{j}$ is zero. It is called basic equation of steady currents.

2. The continuity equation describes the fact that the flux through a closed surface is equal to the decrease of charge in the volume bounded by it. And for steady current, the outward flux is equal to increased flux which makes $\text{div } \vec{j}$ equal to zero.
3. Consider a circuit in which a steady current is maintained. Therefore, there is no source or sink of charge in the circuit and hence the rate of change of charge density ρ with time is zero everywhere.

$$\text{i.e.,} \quad \frac{\partial \rho}{\partial t} = 0$$

and from eqn. (33), we have

$$\vec{\nabla} \cdot \vec{j} = 0$$

But $\vec{j} = \sigma \vec{E}$ where σ is the electrical conductivity. If the conductor is homogeneous, σ is constant everywhere, then

$$\vec{\nabla} \cdot \vec{j} = \vec{\nabla} \cdot (\sigma \vec{E}) = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{E} = 0 \quad \dots(34)$$

If $\vec{E} = \vec{\nabla} \cdot \vec{E}_s + \vec{E}_b$, where $\vec{E}_s$ is the electrostatic component of the electric field and $\vec{E}_b$ is the electric field generated by source of *emf*, then from eqn. (34) we have

$$\vec{\nabla} \cdot \vec{E} = \vec{\nabla} \cdot [\vec{E}_s + \vec{E}_b] = 0$$

$$\text{or} \quad \vec{\nabla} \cdot [\vec{E}_s + \vec{E}_b] = 0$$

In the absence of source of *emf* $\vec{E}_b = 0$, we have

$$\vec{\nabla} \cdot \vec{E}_s = 0$$

But we know that the electrostatic field is related to the potential U as

$$\vec{E}_s = -\vec{\nabla} U$$

$$\therefore \vec{\nabla} \cdot \vec{E}_s = \vec{\nabla} \cdot (-\vec{\nabla} U) = 0$$

$$\Rightarrow -\nabla^2 U = 0$$

$$\text{or} \quad -\nabla^2 U = 0 \quad \dots(35)$$

which is Laplace's equation. Thus, a steady current problem can be solved in the same way as an electrostatic problem, by the application of Laplace's equation.

11.31 ELECTRIC DISPLACEMENT VECTOR D

As the electric field intensity depends upon the medium $\left(E = \frac{q}{4\pi\epsilon r^2}\right)$ we can define another important term **electric displacement vector D** , which depends only on the magnitude of charge and its distribution but is independent of the nature of medium. This vector is given by

$$D = \frac{q}{4\pi r^2}$$

or

$$\vec{D} = \epsilon \left(\frac{1}{4\pi\epsilon} \cdot \frac{q}{r^3} \hat{r} \right) = \epsilon \vec{E} \quad \dots(48)$$

Here vector D is analogous to E and is measured in coulomb/m². The term $D \cdot dA$ gives the flux of normal displacement through surface dA and remains unaltered in any medium.

11.32 DISPLACEMENT CURRENT

This is the current which exists across the capacitor in the circuit when alternating electric field (changing electric field) is applied to the parallel plate capacitor. Because of the presence of vacuum or dielectric material in the capacitor, no charge motion takes place but the displacement of charges takes place inside it and results in the existence of current on the other surface of the capacitor. This current is called displacement current. It is also observed that a magnetic field is set in the region between the plates of the capacitor due to changing electric field although no conduction current flows. Hence, changing electric field across the parallel plate capacitor induces a current known as displacement current which eventually produces magnetic field.

Let q be the charge on the plates of a capacitor having capacitance C and V be the alternating potential applied to the plates. Therefore, the displacement current is given by

$$I = \frac{dq}{dt} \text{ but } q = CV$$

$$\therefore I = C \frac{dV}{dt} = \frac{\epsilon A}{d} \cdot \frac{dV}{dt} \quad \left(\because C = \frac{\epsilon A}{d} \right)$$

Also, $V = Ed$, where E is the electric field and d is the distance between the plates, therefore, we have,

$$I = \frac{\epsilon A}{d} \cdot \frac{d(Ed)}{dt} = \epsilon A \frac{dE}{dt}$$

or

$$\frac{I}{A} = \epsilon \frac{dE}{dt} \Rightarrow J_d = \frac{dD}{dt} \quad (\because D = \epsilon E)$$

where J_d is known as displacement current density.

12.5 MAXWELL'S EQUATIONS

The Maxwell's equations are the basic equations on which the structure of electromagnetism is built. The general equations can be written in terms of the electric and magnetic vectors as follows:

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \dots(4)$$

The eqn. (4) is the statement of Faraday's laws.

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{j} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\text{or } \vec{\nabla} \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t} \quad \dots(5)$$

The eqn. (5) is the Ampere's law which holds for both steady as well as varying currents.

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\text{or } \vec{\nabla} \cdot \vec{D} = \rho \quad \dots(6)$$

The eqn. (6) is the statement of Gauss's law and is true for both static and varying fields, i.e., both for stationary and moving charges.

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \dots(7)$$

The eqn. (7) is a general equation for magnetic fields, which states that the total magnetic flux over any closed surface is zero, i.e., magnetic monopoles do not exist.

These four equations are called the differential forms of the Maxwell's equations.

12.5.1 Derivation of the Maxwell's First Equation

$$\vec{\nabla} \times \vec{E} = \frac{-\partial \vec{B}}{\partial t}$$

This equation can be derived from the Faraday's law of electromagnetism, which states that the induced emf e is given by the time rate of change of magnetic flux $\phi_m = \iint \vec{B} \cdot \vec{ds}$. That is

$$e = - \frac{d\phi_m}{dt} = - \frac{d}{dt} \left[\iint \vec{B} \cdot \vec{ds} \right]$$

$$\text{or } e = \iint \frac{\partial \vec{B}}{\partial t} \cdot \vec{ds}$$

If $\vec{E}$ be the electric field strength, then

$$e = \oint \vec{E} \cdot d\vec{l}$$

$$\therefore \oint \vec{E} \cdot d\vec{l} = - \iint \frac{\partial \vec{B}}{\partial t} \cdot \vec{ds} \quad \dots(8)$$

Now, according to the Stoke's theorem

$$\oint \vec{E} \cdot d\vec{l} = \iint \vec{\nabla} \times \vec{E} \cdot d\vec{s}$$

On substituting above equation in eqn. (23), we get

$$\iint \vec{\nabla} \times \vec{E} \cdot d\vec{s} = - \iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

$$\Rightarrow \vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

DO YOU KNOW?

Maxwells equations are also known as electromagnetic equations.

12.5.2 Derivation of the Maxwell's Second Equation

$$\vec{\nabla} \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$$

This equation is derived from the Ampere's rule. It states that the magnetomotive force F_m is given by the time rate of flow of charge. That is

$$F_m = \frac{dq}{dt} \text{ or } F_m = I$$

where I is the conduction current in circuit (current due to actual motion of the charged particle around the loop). Now, if $\vec{H}$ be the magnetic field, then $F_m = \oint \vec{H} \cdot d\vec{l}$.

Also, if $\vec{j}$ be the current density then

$$I = \iint \vec{j} \cdot d\vec{s}$$

Hence, the Ampere's law can be written as $\oint \vec{H} \cdot d\vec{l} = \iint \vec{j} \cdot d\vec{s}$

Maxwell showed that not only the flow of charges but also changing electric field can cause magnetic field. (Faraday's law states that changing magnetic field cause electric field, so Maxwell's assertion is just the reverse of the Faraday's law of electromagnetic induction.) Maxwell proved that the contribution

of magnetic field due to changing electric field is equivalent to $\frac{\partial \vec{D}}{\partial t}$. It is called displacement current density. So the Ampere's Law is modified to

$$\oint \vec{H} \cdot d\vec{l} = \iint \left(\vec{j} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s} \quad \dots(9)$$

Now, according to Stoke's theorem

$$\oint \vec{H} \cdot d\vec{l} = \iint \vec{\nabla} \times \vec{H} \cdot d\vec{s}$$

On substituting above equation in eqn. (24), we get

$$\iint \vec{\nabla} \times \vec{H} \cdot d\vec{s} = \iint \left(\vec{j} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$$

DO YOU KNOW?

A changing electric field induces a currents known as displacement current.

$$\Rightarrow \vec{\nabla} \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$$

12.5.3 Derivation of the Maxwell's Third Equation

$$\vec{\nabla} \cdot \vec{D} = \rho$$

According to the Gauss's law in electrostatics, the total electric flux through the Gaussian surface is equal to the charge enclosed by it. That is

$$\oiint \vec{D} \cdot \vec{ds} = q$$

Now, if ρ be the volume charge density, then

$$q = \iiint \rho \, dv$$

$$\therefore \oiint \vec{D} \cdot \vec{ds} = \iiint \rho \, dv$$

According to Gauss's divergence theorem

$$\oiint \vec{D} \cdot \vec{ds} = \iiint \vec{\nabla} \cdot \vec{D} \, dv$$

On substituting above equation in eqn. (25), we get

$$\iiint \vec{\nabla} \cdot \vec{D} \, dv = \iiint \rho \, dv$$

$$\Rightarrow \vec{\nabla} \cdot \vec{D} = \rho$$

12.5.4 Derivation of the Maxwell's Fourth Equation

$$\vec{\nabla} \cdot \vec{B} = 0$$

This relation is derived from the magnetic counter part of the Gauss's law in electrostatics. The magnetic lines of force are continuous and do not appear to have the origin or the end. That is, the total magnetic flux through any gaussian surface drawn in the magnetic field is zero. *i.e.*,

$$\oiint \vec{B} \cdot \vec{ds} = 0 \quad \dots(10)$$

According to Gauss's divergence theorem

$$\oiint \vec{B} \cdot \vec{ds} = \iiint \vec{\nabla} \cdot \vec{B} \, dv$$

$\therefore$ On substituting above equation in eqn. (10), we get

$$\iiint \vec{\nabla} \cdot \vec{B} \, dv = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{B} = 0$$

DO YOU KNOW?

Magnetic monopoles do not exist.

12.6 DIFFERENT FORMS OF MAXWELL'S EQUATIONS

The general forms of the Maxwell's equations obtained earlier get modified under different conditions. The change occur due to the following reasons.

1. In free space or vacuum, there is no flow of charge. Also $\vec{j} = 0$. As there is no charged particle present in the space, hence $\rho = 0$.
2. In the presence of steady current alone $\frac{\partial \vec{D}}{\partial t} = 0$. Also $\frac{\partial \vec{B}}{\partial t} = 0$. Because both the electric as well as the magnetic fields remain constant.
3. In the absence of current but in the presence of static charges $\frac{\partial \vec{B}}{\partial t} = 0$, $\vec{j} = 0$ and $\frac{\partial \vec{D}}{\partial t} = 0$.

The Table 12.1 gives the Maxwell's equation under different conditions.

Table 12.1 Maxwell's Equation

<i>Differential Form</i>	<i>Integral Form</i>	<i>Free space</i>	<i>Presence of steady current</i>	<i>Presence of static current</i>
1. $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot d\vec{l} = \iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\vec{\nabla} \times \vec{E} = 0$	$\vec{\nabla} \times \vec{E} = 0$
2. $\vec{\nabla} \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$	$\oint \vec{H} \cdot d\vec{l} = \iint \left(\vec{j} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$	$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$	$\vec{\nabla} \times \vec{H} = \vec{j}$	$\vec{\nabla} \times \vec{H} = 0$
3. $\vec{\nabla} \cdot \vec{D} = \rho$	$\oiint \vec{D} \cdot d\vec{s} = \iiint \rho \, dv$	$\vec{\nabla} \cdot \vec{D} = 0$	$\vec{\nabla} \cdot \vec{D} = \rho$	$\vec{\nabla} \cdot \vec{D} = \rho$
4. $\vec{\nabla} \cdot \vec{B} = 0$	$\oiint \vec{B} \cdot d\vec{s} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$

12.7 IMPORTANCE OF THE MAXWELL'S EQUATIONS

The Maxwell's equations in electromagnetism carry the same significance as the Newton's law of motion have in mechanics and three laws have in thermodynamics. In the development of these equations, Maxwell drew heavily on circuit relations and Faraday's experimental studies in electromagnetism. The main significance of these equations lies in the fact that the conclusions drawn on the basis of these have found excellent agreement with the experimental observations. Also, whereas Newton's laws of motion need modification in view of the Einstein's special theory of relativity; the Maxwell's equations require no change. That is the Maxwell's equations are invariant under the Lorentz transformations. The scope of these equations is so vast that it extends from elementary principles to the large scale electromagnetic devices.

1. Maxwell's First Equation $\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$

(a) It relates the electric field $\vec{E}$ and magnetic induction $\vec{B}$.

(b) It is a time dependent or time varying equation, i.e. $\vec{E}$ is generated by time variation of $\vec{B}$.

(c) It summarizes the well known **Faraday's laws or Lenz's law** of electromagnetic induction.

(d) It relates the space variation of $\vec{E}$ with the time variation of $\vec{B}$ or $\vec{H}$.

2. Maxwell's Second Equation $\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{j} + \frac{\partial \vec{D}}{\partial t} \right)$

(a) It relates the magnetic field $\vec{B}$ with electric displacement vector $\vec{D}$ and the current density $\vec{j}$.

(b) It is also a time dependent equation.

(c) It represents **Ampere's law**.

(d) $\vec{B}$ can be generated from $\vec{j}$ and the time variation of $\vec{D}$ jointly as well as independently.

(e) It relates the space variation of $\vec{B}$ with the time variation of $\vec{D}$ or $\vec{E}$.

3. Maxwell's Third Equation; $\vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0$

(a) It represents **Gauss's Law**.

(b) It is time independent or steady state equation.

(c) The flux of the lines of electric force depends upon charge density.

(d) Charge acts as a source or sink for the lines of electric force.

4. Maxwell's Fourth Equation $\vec{\nabla} \cdot \vec{B} = 0$

(a) It expresses, a well known observation not contradicted by the experiment so far, that *isolated magnetic poles do not exist*.

(b) It states that equal number of lines of magnetic induction enter and leave a given volume.

(c) There is no source or sink for lines of magnetic force.

(d) It is a time independent equation.

DO YOU KNOW?

Maxwell's equations do not need any change in view of Einstein's special theory of relativity.